AI FLOODSHIELD PAK · CHENAB AT MARALA

Model accuracy, class F1, and confusion matrices

Holdout scores for the seven selected horizon models. Training days are monsoon seasons 2024–2025. The test set is monsoon 2026. Accuracy was not the rule used to pick a model.

98.60%
Pooled accuracy, 635 of 644 test days

49.64%
Mean macro-F1, Day +1 through Day +6

0.000
Low-class F1 on every day that has a Low example

How to read the high accuracy. The 2026 labels are almost all Normal. Days +1 to +6 each contain one Low day and no Medium, High, Very High, or Exceptional day. Every selected model missed that Low day, so flood-class F1 is zero even while overall accuracy stays near 99%. Day +7 has no minority class in the test set, so its 99.42% macro-F1 is Normal only. Very High has no historical examples. Exceptional has one, which is below the two-example minimum, so neither class was trained and neither has an F1.

Selected model by horizon

HORIZON	MODEL	TEST ROWS	CORRECT	ACCURACY	MACRO-F1	CLASSES IN MACRO-F1
Day +1	Random Forest, unweighted	97	95	97.94%	0.495	Normal, Low
Day +2	Majority baseline	95	94	98.95%	0.497	Normal, Low
Day +3	Random Forest, balanced	94	92	97.87%	0.495	Normal, Low
Day +4	Random Forest, unweighted	93	92	98.92%	0.497	Normal, Low
Day +5	Random Forest, unweighted	90	89	98.89%	0.497	Normal, Low
Day +6	Majority baseline	88	87	98.86%	0.497	Normal, Low
Day +7	Random Forest, balanced	87	86	98.85%	0.994	Normal only

Pooled accuracy is 635 correct predictions divided by 644 test rows. Macro-F1 averages F1 only over classes present in that horizon's 2026 test set.

F1 score for each class

F1 is the harmonic mean of precision and recall. A dash means the class has no score: it is missing from the 2026 test set, or it was unsupported in training.

HORIZON	NORMAL	LOW	MEDIUM	HIGH	VERY HIGH	EXCEPTIONAL
Day +1	0.990	0.000	—	—	—	—
Day +2	0.995	0.000	—	—	—	—
Day +3	0.989	0.000	—	—	—	—

HORIZON	NORMAL	LOW	MEDIUM	HIGH	VERY HIGH	EXCEPTIONAL
Day +4	0.995	0.000	—	—	—	—
Day +5	0.994	0.000	—	—	—	—
Day +6	0.994	0.000	—	—	—	—
Day +7	0.994	—	—	—	—	—

Exact scored values: Day +1 Normal 0.9896; Day +2 Normal 0.9947; Day +3 Normal 0.9892; Day +4 Normal 0.9946; Day +5 Normal 0.9944; Day +6 Normal 0.9943; Day +7 Normal 0.9942. Low F1 is exactly 0 wherever a Low day exists. Day +7 predicted Low once, but there was no true Low day, so Low F1 is not defined.

Confusion matrices

Rows are the true 2026 label. Columns are the prediction. A shaded diagonal cell is a correct call. Class order is Normal, Low, Medium, High, Very High, Exceptional. Every Medium, High, Very High, and Exceptional cell is zero.

Day +1 · Random Forest, unweighted

95 correct of 97 · accuracy 97.94%

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**95**	1	0	0	0	0
Low	1	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Day +2 · Majority baseline

94 correct of 95 · accuracy 98.95%

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**94**	0	0	0	0	0
Low	1	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Day +3 · Random Forest, balanced

92 correct of 94 · accuracy 97.87%

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**92**	1	0	0	0	0
Low	1	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Day +4 · Random Forest, unweighted

92 correct of 93 · accuracy 98.92%

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**92**	0	0	0	0	0
Low	1	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Day +5 · Random Forest, unweighted

89 correct of 90 · accuracy 98.89%

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**89**	0	0	0	0	0
Low	1	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Day +6 · Majority baseline

87 correct of 88 · accuracy 98.86%

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**87**	0	0	0	0	0
Low	1	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Day +7 · Random Forest, balanced

86 correct of 87 · accuracy 98.85%. No true Low day; one Normal day was called Low.

TRUE \ PRED	NORMAL	LOW	MED	HIGH	V.HIGH	EXC.
Normal	**86**	1	0	0	0	0
Low	0	**0**	0	0	0	0
Medium	0	0	**0**	0	0	0
High	0	0	0	**0**	0	0
Very High	0	0	0	0	**0**	0
Exceptional	0	0	0	0	0	**0**

Test-set support

HORIZON	NORMAL	LOW	MEDIUM	HIGH	VERY HIGH	EXCEPTIONAL
Day +1	96	1	0	0	0	0
Day +2	94	1	0	0	0	0
Day +3	93	1	0	0	0	0
Day +4	92	1	0	0	0	0
Day +5	89	1	0	0	0	0
Day +6	87	1	0	0	0	0
Day +7	87	0	0	0	0	0

Counts of true labels in the 2026 test window. Full historical inflow labels: Normal 316, Low 14, Medium 3, High 2, Very High 0, Exceptional 1.

Source: models/metrics/horizon_metrics.json, version floodshield-marala-7day-2. Report dated 22 October 2026. Figures are the saved 2026 evaluation of the selected model for each horizon.